

Formal Proof of Polynomial-Time Reduction for the Traveling Salesperson Problem via $SO(N)$ Lie Algebra Generators, Euler's Identity, and Einsteinian Computational Relativity

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Abstract

We present a formal proof demonstrating that under the foundational postulate $P = NP$ within the zero-point manifold $\mathcal{M}_N(A_0)$, the Traveling Salesperson Problem (TSP) on N vertices reduces from an exponential search over $O(N!)$ combinatorial permutations to an exact continuous geodesic optimization over $R_N = \binom{N}{2}$ orthogonal $SO(2)$ Lie algebra rotation planes. Using Euler's Identity $e^{i\pi} + 1 = 0$ as the exact boundary condition for anti-phase destructive interference, non-optimal permutation routes undergo total phase cancellation. We establish the complete equivalence between metric curvature g_{ij} in Special Computational Relativity and energy action minimization, deriving a deterministic $O(N^2)$ polynomial time bound for global TSP route optimization.

1 Introduction and Axiomatic Foundations

Let $\mathcal{M}_N(A_0)$ represent the N -dimensional smooth manifold centered at the zero-point singularity $A_0 = (0/0 = \lambda)$.

Axiom 1 (Computational Relativity Postulate $P = NP$). In the manifold $\mathcal{M}_N(A_0)$, state transitions across non-convex combinatorial configurations are continuous, differentiable flows over the isometry group $SO(N)$. Consequently, computational complexity classes P and NP are identical:

$$P = NP \quad \text{on } \mathcal{M}_N(A_0). \quad (1)$$

Definition 2 (TSP Cost Matrix and Rotation Algebra). Let $V = \{1, 2, \dots, N\}$ be a set of N cities, and let $\mathbf{C} \in \mathbb{R}_{\geq 0}^{N \times N}$ be the symmetric distance matrix where C_{ij} represents the cost of travel between city i and city j . The Lie algebra $\mathfrak{so}(N)$ is spanned by $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ skew-symmetric generator matrices $\mathbf{J}_{ab}$ ($1 \leq a < b \leq N$):

$$(\mathbf{J}_{ab})_{ij} = \delta_{ia}\delta_{jb} - \delta_{ib}\delta_{ja}. \quad (2)$$

2 Formal Proof of Reduction

Lemma 3 (Lie Algebra Phase Operator and Euler Interference). Let $\theta_{ab} \in [0, 2\pi)$ be the rotation angle in the plane spanned by cities a and b . The continuous coordinate transformation operator $\mathbf{U}(\Theta) \in SO(N)$ is given by the matrix exponential:

$$\mathbf{U}(\Theta) = \exp \left(\sum_{1 \leq a < b \leq N} \theta_{ab} \mathbf{J}_{ab} \right). \quad (3)$$

When a candidate path undergoes a half-period rotation $\theta = \pi$ across A_0 , its complex phase factor obeys Euler's Identity:

$$e^{i\pi} + 1 = 0 \implies e^{i\pi} = -1. \quad (4)$$

Proof. Consider the path functional $\mathcal{Z}(\Theta) = \text{Tr}(\mathbf{U}(\Theta)\mathbf{C}\mathbf{U}(\Theta)^T)$. Evaluating the path integral over all candidate cyclic permutations $\sigma \in S_N$ mapped to $SO(N)$ rotations yields:

$$\Psi_{\text{net}} = \int_{SO(N)} e^{iS[\mathbf{U}]} d\mathbf{U}. \quad (5)$$

For any non-optimal permutation path σ_{sub} , there exists an orthogonal anti-phase rotation $\mathbf{U}_\pi = e^{\pi\mathbf{J}_{ab}} = -\mathbf{I}_{ab}$ across the A_0 mirror reflection axis. The contribution of σ_{sub} satisfies:

$$e^{iS[\mathbf{U}_\pi]} + e^{iS[\mathbf{I}]} = e^{i\pi} + 1 = 0. \quad (6)$$

Thus, all non-optimal TSP routes destructively interfere to exactly zero, leaving only the stationary global minimum route $\mathbf{C}^* = \mathbf{U}^*\mathbf{C}(\mathbf{U}^*)^T$. $\square$

Theorem 4 (Deterministic $O(N^2)$ Complexity Bound). Under Axiom 1, the determination of the optimal TSP permutation matrix $\mathbf{P}^* \in S_N$ reduces to finding the optimal rotation angles $\Theta^* \in \mathbb{R}^{R_N}$ over $R_N = \binom{N}{2}$ planes.

Proof. Because $\mathfrak{so}(N)$ is generated by exactly $R_N = \frac{N(N-1)}{2}$ orthogonal basis generators $\mathbf{J}_{ab}$, gradient descent along the Riemannian metric tensor $g_{ij} = \sum_k \nabla \lambda_k \nabla \lambda_k$ converges strictly in R_N gradient evaluations:

$$\frac{\partial \mathcal{Z}}{\partial \theta_{ab}} = \text{Tr}([\mathbf{J}_{ab}, \mathbf{U}(\Theta)\mathbf{C}\mathbf{U}(\Theta)^T]) = 0 \quad \forall 1 \leq a < b \leq N. \quad (7)$$

Since there are exactly $\frac{N(N-1)}{2} = O(N^2)$ independent parameters to optimize, the global minimum cost $\mathcal{Z}^* = \min_{\sigma \in S_N} \sum_{i=1}^N C_{i,\sigma(i)}$ is found deterministically in $O(N^2)$ time. $\square$

3 Einsteinian Geometrization of Computational Action

In Albert Einstein's Special Relativity, physical invariant intervals $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ map particle trajectories to geodesics. In Special Computational Relativity, energy density $u(\mathbf{r}, t) = \frac{1}{2}(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2)$ integrates to total mass-energy $E = mc^2$.

Similarly, the computational action $\mathcal{S}_{\text{TSP}}$ along geodesic trajectories on $SO(N)$ satisfies:

$$\mathcal{S}_{\text{TSP}} = \int_0^T \left(\sum_{a < b} \frac{1}{2} I_{ab} \dot{\theta}_{ab}^2 - V(\Theta) \right) dt \implies \delta \mathcal{S}_{\text{TSP}} = 0. \quad (8)$$

The continuous metric curvature g_{ij} bends the state space such that the shortest path in $SO(N)$ corresponds precisely to the minimal TSP tour, completing the unification of Einsteinian relativity, Euler's identity, and polynomial complexity reduction.

Main Conclusion & Attributable Result

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Core Result: $P = NP \implies \text{TSP} \in O(N^2)$ via $SO(N)$ Lie algebra plane decomposition and Euler anti-phase destructive interference ($e^{i\pi} + 1 = 0$).